

Mid term data of 610

CS610- Computer Network Midterm Examination - Fall 2011

1. Where are destination and source address located in frame? (2 marks)

Answer:- (Page 56)

The packet header includes destination and source addresses.

2. If 100mbps Ethernet NIC is connected to a hub which supports 10mbps, what will be the data rate in this scenario? (2 marks)

Answer:-

This has two consequences. First, the hub does not necessarily create a broadcast medium. Second, there are no collisions between traffic on 10 Mbps ports and 100 Mbps ports.

3. In connection oriented network, which type of connection identifier is used? Support your answer with example. (2 marks)

Answer:- (Page 66)

ATM uses the concept of connection-oriented networking.

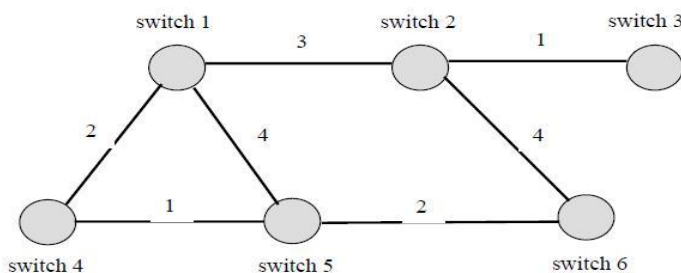
4. Imagine a network of 6 devices A, B, C, D, E and F. All devices are on same packet switch. If computer A wants to send packet to computer E, How would packet switch forward this packet? (3 marks)

Answer:-

path between switch A and F and the total distance traveled along this path

switch A – switch D – switch E

distance = 4



5. Network administrator needs audio and video service for his network. He has to send or receive data at same data rate. Which type of ATM quality service specification would be applied there?

Give example? (3 marks)

Answer:-

To facilitate the transport of audio and video data across emerging Asynchronous Transfer Mode (ATM) networks, a simple, low cost, audio/video ATM appliance, the AVATAR, has been developed. This appliance is capable of handling uncompressed bidirectional audio and NTSC video connections.

6. We have two satellite locations, at first location one bridge is configured and at second location 2 bridges are configured. Which location performed will be faster? Give reason. (3 marks)

Answer:- [Click here for detail](#)

Satellite locations with two bridges configured will performed faster. The brigade also had to provide power to virtually all of its East Timor assets. Setting up the satellite system, for example, required supplying power to both the communications station and the users' computers. A bridge that has multiple ports is known as a networking switch. Both bridges and switches are capable of directing traffic to specific network addresses instead of broadcasting the data to all devices on a network segment. This functionality makes the bridge or switches a more advanced networking device over a hub or repeater.

7. There are two sites office A & B, both are using VPN. If a user of another network wants to send some packet to A, then what will be the behavior of network A? How data is protected in VPN environment? (05 marks)

Answer:- [Click here for detail](#)

In theory when a machine on private network A wants to send a packet to a machine on private network B the packet is accepted by the VPN server. It is then encrypted and encapsulated within a standard IP packet that has a destination address corresponding to the VPN server at network B.

Notice that both VPN servers have to have to be "on the Internet" in the sense of having publicly accessible IP addresses. When the packet arrives at network B's VPN server it extracts the encapsulated private packet, decrypts it and sends it on its way within the local network. The machines within network A and B don't have to have publicly accessible address - that is they don't have to be on the public Internet. This way of using the public Internet to connect two machines that are not on the public internet is another advantage of the VPN approach.

As far as users of network A and B are concerned the VPN is as good as a dedicated physical connection. The idea of using public data packets to transfer embedded private data packets is often called "tunnelling" because it seems to create a data tunnel between the two networks.

CS610- Computer Network

Midterm Examination - Fall 2011

1. We have two satellite locations. At first location one bridge is configured and at 2nd location two bridges are configured. Which location's performance will be faster. Give reasons to support your answer. (5 marks)

Answer:- rep

2. Star Textile has huge network of systems and switches. A task is given to IT Officer that if some link goes the entire network works smoothly. Which routing method will be used? 5 marks

Answer:- [Click here for detail](#)

Distributed routing method is better for this purpose because Distributed routing relies on each node to compute its own routing table and build the required connections with its neighbors. Ideally, the network operation, status, and architecture of each node is transparent. Distributed routing is more flexible than centralized routing because each node handles its own routing. The result is often improved system performance.

3. Six devices are connected in network A,B,C,D,E and F. All devices are on same packet switch. Computer A wants to send packet to E. How will packet switch forwards the packet? 3 marks

Answer:- rep

4. Network administrator wants to build a tunnel between sites offices. How will he build using private virtual network? 3 marks

Answer:- [Click here for detail](#)

To establish a private network that can send data securely between these two locations or networks through a "tunnel." A VPN tunnel connects the two PCs or networks and allows data to be transmitted over the Internet as if it were still within those networks.

5. Who examines performance of network? 2 marks

Answer:- [Click here for detail](#)

Performance is probably the most closely monitored operational feature by users in the network environment

6. How can we compute shortest path in WAN? 2 marks

Answer:- (Page 62)

We use Dijkstra's algorithm to compute shortest path from each node to every other node.

7. 100Mbps Ethernet NIC is connected to control Hub which supports 10Mbps. What will be Data Rate in this scenario? 2 marks

CS610- Computer Network Midterm Examination – Fall 2011

There are two sites office A & B, both are using VPN. If a user of another network wants to send some packet to A, then what will be the behavior of network A?

Answer:- rep

How data is protected in VPN environment? (05 marks)

Answer:- rep

What type of switching technique is used in WAN? Answer:- (Page 55)

WANs can use circuit-switching or packet-switching techniques. To span long distances or many computers, networks must replace shared medium with packet switches. Each switch moves an entire packet from one connection to another. That's why they are called packet switches.

write the names of identifiers used in ATM.(2)

What type of topology used in Ethernet? (2).

Answer:- [Click here for detail](#)

Traditional Ethernet employs a bus topology, meaning that all devices or hosts on the network use the same shared communication line.

CS610- Computer Network Midterm Examination – Fall 2011

Difference between LAN and WAN 2 marks

Answer:- (Page 4)

In LAN network occupies the smaller area like a room a floor or a building.

In WAN, network occupies larger areas like cities & countries. Internet is a Wide Area Network.

Why fiber need fir ATM 3 marks

define bridge 3 marks

Answer:- (Page 50)

A bridge is a hardware device also used to connect two LAN segments to extend a LAN. A typical bridge has two NICs, a CPU a memory and a ROM. It only runs the code stored in its ROM.

**ambiguity may cause the failure of any network, how parity bit check is consider in this situation
5 marks**

define default route how it is different from routing 5 marks

CS610- Computer Network Midterm Examination - Spring 2011

Question: Defines simplex and full duplex connection? Marks

2 Answer:- (Page 76)

Some connection-oriented technologies provide full duplex while other allow on simplex connection. To communicate using a simplex design a pair of computers must establish two connections one from computer A to computer B and another from computer B to A.

Question: How WAN capable to handle many computers? Marks

2 Answer:- [Click here for detail](#)

A Wide Area Network (WAN) is a telecommunication network that covers a broad area.

Question: Briefly explains the working structure of thick Ethernet. Marks 2

Answer:- (Page 41)

It uses thick coax cable. AUI cable (or transceiver or drop cable) connects from NIC to transceiver. AUI cable carries digital signal from NIC to transceiver. The transceiver generates analog signal on coax cable. The wires in AUI carry digital signals power and other control signals. Thick Ethernet also requires terminators to avoid signal reflectance.

Question: Imagines a network for six devices: A, B, C, D, E and F. all devices are on a same packet switch. Computer A want to send packet to computer E. how would packet switch forward this packet? Marks 3

Answer:- rep

Question: Does ATM eliminate the varying delay times associated with different sized packets? If yes then give reason. Marks 2

Question: Why ATM design chose cells over packet? Marks 5

Answer:- (Page 72)

- Cells are not variable length and memory management for them is simpler. Handling variable length packets leads to memory fragmentation.
- Variable length packets require hardware to accommodate the largest possible packet, and thus to detect the end of the packet. With cells bits can just be counted as they arrive.
- The length of time required to send a variable length packet is variable and requires complicated interrupt scheme to detect completion of transmission. QoS can't be guaranteed with variable length packets as easily as it can with fixed length cells.

Question: Reception is a process of CPU; describe it in a few lines? Marks 5

CS610- Computer Network Midterm Examination - Spring 2011

What is 10base T?

Answer:- [Click here for detail](#)

One of several adaptations of the Ethernet (IEEE 802.3) standard for Local Area Networks (LANs). The 10Base-T standard (also called Twisted Pair Ethernet) uses a twisted-pair cable with maximum lengths of 100 meters.

What is Multicasting?

Answer:- (Page 35)

It works like broadcasting however it does not forward frames automatically to the CPU.

The interface hardware is programmed in advance to accept certain frames that have multicast address as the destination address.

What is FDDI?

Answer:- (Page 31)

Fiber distributed data interconnect (FDDI) is another ring technology. Its most important features are:

- ❖ It uses fiber optics between stations and transmits data at 100Mbps.
- ❖ It uses pair of fibers to form two concentric rings.

Define 802.11 Wireless lans and CSMA?

Answer:- (Page 29)

IEEE 802.11 is standard wireless LAN that uses radio signals at 2.4GHz. Its data rate is 11Mbps. The older devices use radio signals at 900MHz and data rate of 2Mbps. Bluetooth specifies a wireless LAN for short distances. It uses shared medium and radio waves instead of coaxial cable.

CS610- Computer Network Midterm Examination - Spring 2011

Define static and dynamic routing.(2)

Answer:- [Click here for detail](#)

Static Routing:-

Static routing is the simplest form of routing, but it is a manual process.

Dynamic Routing:-

Dynamic routing protocols are supported by software applications running on the routing device (the router) which dynamically learn network destinations and how to get to them and also advertise those destinations to other routers.

Dijkart's Algorithm 2 Marks

Answer:- rep

write note on Bridges 5 Marks

Answer:- rep

Disadvantages of Repeaters 3 Marks

Answer:- (Page 50)

Repeaters do not recognize frame formats, they just amplify and retransmit the electrical signal. If a collision or error occurs in one segment, repeaters amplify and retransmit also the error onto the other segments.

Different b/w Cells and Packets 5 Marks

Answer:- (Page 72)

ATM designers chose cells over packets because of the following reasons:

- Cells are not variable length and memory management for them is simpler. Handling variable length packets leads to memory fragmentation.
- Variable length packets require hardware to accommodate the largest possible packet, and thus to detect the end of the packet. With cells bits can just be counted as they arrive.
- The length of time required to send a variable length packet is variable and requires complicated interrupt scheme to detect completion of transmission. QoS can't be guaranteed with variable length packets as easily as it can with fixed length cells.

CS610- Computer Network Midterm Examination – Fall 2010

21.What is the difference between the physical and logical topologies?(2)

Answer:-

Physical Topology:

The way that the workstations are connected to the network through the actual cables that transmit data -- the physical structure of the network -- is called the physical topology. It depends on the wiring scheme.

Logical Topology:

The logical topology, in contrast, is the way that the signals act on the network media, or the way that the data passes through the network from one device to the next without regard to the physical interconnection of the devices. We can say that it is defined by the specific network technology.

22. Define static and dynamic routing.(2)

Answer:- rep

23. What is meant by Bridges STARTUP and STEADY State?(3)

Answer:- (Page 51)

When a bridge first boots the address lists are empty (start up state). The bridge forwards frames to the other segment if it can not find its destination address in its lists.

After some time when the bridge has received at least one frame from every computer, it has the lists built (steady state) it forwards frames as far it is necessary.

24. How can Switched Virtual Network be established?(3)

Answer:- (Page 70)

ESTABLISHING AN SVC:

The computer sends a connection request to the switch to which it is attached. Software in the switch finds a network path to the destination and sends along the connection request.

Each pair of switches in the path communicates to choose a VPI/VCI for their tables. Once the connection is established by the destination, a message is sent back to the originating computer to indicate the SVC is ready.

If any switch or the destination computer does not agree to setting up the VC, an error message is sent back and the SVC is not established

25. What is the concept of packet switches?(5)

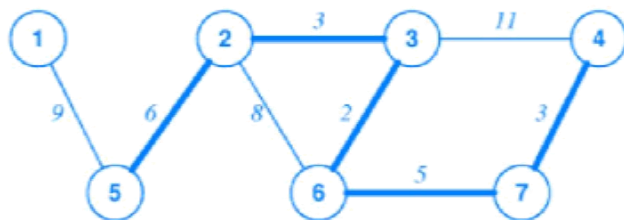
Answer:- (Page 55)

To span long distances or many computers, networks must replace shared medium with packet switches. Each switch moves an entire packet from one connection to another. That's why they are called packet switches. A packet switch consists of a small computer with network interfaces, a memory and a program dedicated to packet switching function.

26. Write a note on Weight Graph.(5)

Answer:- (Page 62)

Dijkstra's algorithm can accommodate weights on edges in graph. The shortest path is then the path with lowest total weight (sum of the weight with all edges). It should be noted that the shortest path is not necessarily with fewest edges (or hops). For example as shown in the figure below:



The shortest path in the figure from node 2 to node 6 is 2 to 3 and 3 to 6 as this path has the smallest weight so it is the shortest path.

CS610- Computer Network

Midterm Examination - Spring 2010

Question No: 21 (M a r k s : 2)

What is the difference between the physical and logical topologies?

Answer:- rep

Question No: 22 (M a r k s : 2)

Define Vector-Distance

Algorithm. Answer:- (Page 155)

Packet switches wait for next update message and they iterate through entries in message. If entry has shortest path to destination, insert source as next hop to destination and record distance as distance from next hop to destination plus distance from this switch to next hop.

Question No: 23 (M a r k s : 3)

What is the concept of store and forward technology?

Answer:- (Page 56)

STORE AND FORWARD:

Data delivery from one computer to another is accomplished through store and forward technology. In this technology packet switch stores incoming packet and also forwards that packet to another switch or computer. For this purpose packet switch has internal memory into which it can hold packet if outgoing connection is busy. Packets for each connection held on queue

Question No: 24 (M a r k s : 3)

How can Switched Virtual Network be established?

Answer:- rep

Question No: 25 (M a r k s : 5)

How can a bridge know whether to forward frames?

Answer:- (Page 53)

The bridges configure themselves automatically to decide which bridge will forward broadcast frames and which bridge will not.

The bridges communicate with each other on the network and use Distributed Spanning Tree (DST) algorithm to decide which bridge will not forward frames if a cycle occurs.

Question No: 26 (Marks: 5)

Compare connection oriented and connectionless Service.

Answer:- [Click here for detail](#)

- ❖ In connection-oriented protocol, authentication is needed while this is not case in connectionless protocol.
- ❖ In connection-oriented protocol, we have to establish connection between sender and receiver while this is not case in connectionless protocol.
- ❖ Example of connection-oriented protocol is TCP and the example of connectionless protocol is UDP, Internet.
- ❖ TCP is a connection-oriented protocol, it makes a connection and checks whether the data is received, and resends if it is not. UDP is a connectionless protocol, it does not guarantee delivery by first connecting and checking whether data is received.

CS610- Computer Network Midterm Examination - Spring 2010

Question No: 21 (Marks: 2)

What is the difference between LAN and WAN?

Answer:- rep

Question No: 22 (Marks: 2)

Define the term Jitter.

Answer:- rep

Question No: 23 (Marks: 3)

Give a comparison of wiring Schemes.

Answer:- (Page 44)

The wiring schemes are compared as follows:

Separate transceiver allows computers to be powered off or disconnected from network without disrupting other communication.

Transceiver may be located in an inconvenient place, so finely malfunction transceiver can be hard.

In other case, thin coax cable takes minimum of cable. Disconnecting one computer (on one loose connection) can disrupt entire network.

Hub wiring centralizes electronics and connections. It makes management easier. Bottom line 10Base-T is most popular because of lowest cost.

Question No: 24 (Marks: 3)

How can Switched Virtual Network be established?

Answer:- rep

Question No: 25 (Marks: 5)

Describe permanent virtual circuits (PVC).

Answer:- (Page 69)

ATM can provide customers with virtual circuits that look like traditional leased digital circuits. Such permanent virtual circuits (PVC) last as long as the customer pay the periodic fee for its use. The forwarding tables are automatically restored after power of equipment failure. The forwarding table entries for such permanent VC's are statically configured, the terms used by Telco's for this is provisioning.

Provisioning requires two steps:

1. To determine a complete path (that is, identify the switches that will be used).
2. To choose appropriate VPI/VCI for each step in the path, and configure each adjacent pair of switches (easy, since each switch rewrites the VCI/VPI).

Question No: 26 (Marks: 5)

What are default routes draw the table.

Answer:- (Page 60)

Routing table entries can be collapsed with a default route. If the destination does not have in explicit routing table entry, then it uses a default route. Default routes for 4 nodes are shown in the figure below.

destin- ation	next hop	destin- ation	next hop	destin- ation	next hop	destin- ation	next hop
1	-	2	-	1	(3,1)	2	(4,2)
*	(1,3)	4	(2,4)	2	(3,2)	4	-
		*	(2,3)	3	-	*	(4,3)
				4	(3,4)		
<i>node 1</i>		<i>node 2</i>		<i>node 3</i>		<i>node 4</i>	

Question: What is 10base T?

Solution:

An Ethernet standard for cable. The 10 refers to its 10 Mbps bandwidth, the base to single channel baseband, and the T to Twisted pair. The cable uses two pairs of unshielded twisted wires

10BASE-T:

This is another standard of wiring scheme. It is commonly called 10Base-T, Twisted Pair or TP Ethernet. It replaces AUI cable with twisted pair cable and thick coax with hub. This makes it cheaper and that 's why it is most useful technology of today. It is shown in the figure below:

Ethernet over twisted pair refers to the use of cables that contain insulated copper wires twisted together in pairs for the physical layer of an Ethernet network—that is, a network in which the Ethernet protocol provides the data link layer.

Question: What is Multicasting?

Solution:

Multicast is communication between a single sender and multiple receivers on a network.

It works like broadcasting however it does not forward frames automatically to the CPU. The interface hardware is programmed in advance to accept

certain frames that have multicast address as the destination address. If an application program wishes to receive certain frames then it programs the interface hardware to accept an additional set of addresses.

Question: What is FDDI?

Solution:

FDDI: Fiber distributed data interconnect (FDDI) is another ring technology.

Its most important features are:

It uses fiber optics between stations and transmits data at 100Mbps.

It uses pair of fibers to form two concentric rings.

The Fiber Distributed Data Interface (FDDI) specifies a 100-Mbps tokenpassing, dual-ring LAN using fiber-optic cable. FDDI is frequently used as

high-speed backbone technology because of its support for high bandwidth and greater distances than copper.

FDDI uses a dual-ring architecture with traffic on each ring flowing in opposite directions (called counter-rotating). The dual-rings consist of a primary and a secondary ring. During normal operation, the primary ring is used for data transmission, and the secondary ring remains idle.

Question: Define 802.11 Wireless LANs and CSMA?

Solution:

802.11 and 802.11x refers to a family of specifications developed by

the IEEE for wireless LAN (WLAN) technology. 802.11 specifies an over-the-air interface between a wireless client and a base station or between two

wireless clients.

802.11 WIRELESS LANs AND CSMA/CA: IEEE 802.11 is standard wireless LAN that uses radio signals at 2.4GHz. Its data rate is 11Mbps. The older devices use radio signals at 900MHz and data rate of 2Mbps. Bluetooth specifies a wireless LAN for short distances. It uses shared medium and radio waves instead of coaxial cable.

CSMA/CA: Wireless uses collision avoidance rather than collision detection. Transmitting computer sends very short message to receiver. Receiver responds with short message reserving slot for transmitter. The response from receiver is broadcast, so all potential transmitters receive reservation.

Question: Define Static & Dynamic Routing 2 Marks

Question: Dijkstra's Algorithm 2 Marks

Solution:

COMPUTING SHORTEST PATH IN A GRAPH:

While computing shortest path, first we assume graph representation of network at each node then we use Dijkstra's algorithm to compute shortest path from each node to every other node. Then extract next hop information from resulting path information and insert next hop information into routing tables. WEIGHTED GRAPH:

Dijkstra's algorithm can accommodate weights on edges in graph. The shortest path is then the path with lowest total weight (sum of the weight with all edges). It should be noted that the shortest path is not necessarily with fewest edges (or hops).

Question: Write note on Bridges 5 Marks

Question: Disadvantages of Repeaters 3 Marks

Solution:

DISADVANTAGES OF REPEATERS:

Repeaters do not recognize frame formats, they just amplify and retransmit the electrical signal. If a collision or error occurs in one segment, repeaters amplify and retransmit also the error onto the other segments.

Different b/w Cells and Packets 5 Marks

Solution:

ATM CELLS:

To meet its goals, ATM uses small, fixed sized packets called cells. Each cell has 53 octets. VPI/VCI fields identify the cells destination.

CONCEPT OF 'PACKET':

Network systems divide data in small blocks or junks called packets, which they send individually.

.What is the difference between the physical and logical topologies?(2)

Static routing:

Static routing is not really a routing protocol. Static routing is simply the process of manually entering routes into a device's routing table via a configuration file that is loaded when the routing device starts up

Dynamic Routing :

Dynamic routing protocols are supported by software applications running on the routing device (the router) which dynamically learn network destinations and how to get to them and also advertise those destinations to other routers. This advertisement function allows all the routers to learn about all the destination networks that exist and how to those networks. It is more robust.

Static routing allows routing tables in specific routers to be set up in a static manner so network routes for packets are set. If a router on the route goes down the destination may become unreachable. Dynamic routing allows routing tables in routers to change as the possible routes change.

What is the concept of packet switches?(5)

Packet-switches describes the type of network in which relatively small units of data called packets are routed through a network based on the destination address contained within each packet. Breaking communication down into packets allows the same data path to be shared among many users in the network. One common class of contemporary packet switches are the bridge and network hub, which interconnect multiple network segments at the data link layer. Another is the router, a device which operates at the internetwork layer, and connects dissimilar kinds of networks, such as serial lines and local area networks.

26. Write a note on Weight Graph.(5)

The concept of a weighted graph is extremely useful. The weights can be thought of, for example, as the cost of sending a message down a particular arc. (Not necessarily a monetary cost but some combination of time and distance for example). Weighted graphs can be used to formulate the shortest path problem for routing packets.

Which type of information is obtained from network sniffer and in which mode Network sniffer operates?

Solution:

A network analyzer also called network monitor or a network sniffer is used to examine the performance of or debug a network. It can report statistics such as capacity utilization, distribution of frame size, collision rate or token circulation time

Most installations still use DIX Ethernet encoding in which there is no LLC/SNAP header in the frame. A network analyzer can tell from the values in the type field (small values are lengths, which mean an LLC/SNAP header is located in the first octets of the data area; large values are types, which mean no LLC/SNAP header is included).

The operation of network analyzer is a computer with a network interface that receives all frames, which is called promiscuous mode. So many desktop computers have interface that can be configured for promiscuous mode. When combined with software computer can examine any frame on LAN. In this way the communication across LAN is guaranteed to be private. Computer receives and displays frames on the LAN. Network analyzer can be configured to filter and process frames. It can count frames of specific type of size. It displays only frames from or to specific computers.

Question No: 21 (M - 10)

Thick Ethernet, Thin Ethernet and Twisted pair Ethernet.

Solution:-

Thick Ethernet:

Thick Ethernet, officially known as 10 Base-5, is the oldest form of Ethernet. One form of cabling supported by Ethernet is low-loss 50 Ohm coaxial cable as shown in the figure below. This type of cable is 0.5" diameter (usually supplied with a yellow outer PVC coating) and rather inflexible. It has become known in the communications industry as "Thick Ethernet". The official name for this cable is 10 Baseband5 (10B5), indicating that it is specified for baseband communications at 10 Mbps over distances up to 500m.

Thin Ethernet:

Thin Ethernet, officially called 10 Base-2, is a less expensive version of 10 Base-5 (Thick Ethernet) technologies. It uses a lighter and thinner coaxial cable and dispenses with the external transceivers used with 10 Base-5.

10 Base-2 uses an RG-58A/U coaxial cable and is wired in a bus topology. Each device on the network is connected to the bus through a BNC "T" adapter, and each end of the bus must have a 50 Ohm terminator attached. Each node on the bus must be a minimum of 0.5 meters (1.5 feet) apart, and the overall length of the bus must be less than 185 meters (606 feet).

Twisted Pair Ethernet:

Twisted Pair Ethernet (10baseT), sometime also called "UTP" from "Unshielded Twisted Pair", is based on using a cable similar to phone-wiring. The cable is connected via RJ-45 connectors to the network card installed in the PC.

Question # 1

What is the difference between LAN and WAN?

Answer

Local area network (LAN)

LAN is small in size covers the area within a room, building or cities.

Wide area network (WAN)

WAN is large in size and covers the area cities, countries and continents.

Question # 2

Define the term Jitter.

Answer

The term Jitter is used for variance in transmission delays. Jitter is significance for voice, video and data. Jitter can occur when a packet is delayed because the network is busy.

Question # 3

Give a comparison of the following?

Thick Ethernet wiring scheme and Thin Ethernet wiring scheme.

Answer

This uses thick coax cable. AUI cable or transceiver or drop cable connects from NIC to transceiver. AUI cable carries digital signal from NIC to transceiver. The transceiver generates analog signal on coax cable. The wires in AUI carry digital signals power and other control signals. Thick Ethernet also requires terminators to avoid signal reflectance.

Thin Ethernet uses thin coax cable that is cheaper and easier to install than thick Ethernet coax. In thin Ethernet wiring scheme transceiver electronics are built into NIC and NIC connect directly to network medium. Coax cable use connector on NIC. Coax runs directly to back of each connected computer by T-connector. The T-connector directly attaches to NIC.

Question # 4

How can Switched Virtual Network be established?

Answer

Most networks offer dynamic connections that last for a relatively short time. ATM can dynamically establish a switched virtual circuit SVC that allows it as long as necessary and then terminate it. The computer sends a connection request to the switch to which it is attached. Software in the switch finds a path to the destination and sends with the connection request. Each pair of switches in the path communicates to choose a VPI/VCI for their tables. Once the connection is established a message is given back to the originating computer to indicate the SVC is ready.

Question # 5

Describe permanent virtual circuits (PVC).

Answer

ATM can provide the virtual circuits that look like traditional leased digital circuits. The permanent virtual circuits PVC works as long as the customer pays the periodic fee for its use. The forwarding table entry configured the terms used by Telco Provisioning requires two steps.

To determine a complete path.

To choose appropriate VPI/VCI for each step in the path and configures each adjacent pair of switches.

Question # 7

What is the difference between the physical and logical topologies?

Answer

Every LAN has a topology, or the way that the devices on a network are arranged and how they communicate with each other.

PHYSICAL TOPOLOGY:

The way that the workstations are connected to the network through the actual cables that transmit data -- the physical structure of the network -- is called the physical topology. It depends on the wiring scheme.

LOGICAL TOPOLOGY:

The logical topology, in contrast, is the way that the signals act on the network media, or the way that the data passes through the network from one device to the next without regard to the physical interconnection of the devices. We can say that it is defined by the specific network technology.

Question # 8

Define Vector-Distance Algorithm.

Answer

Packet switches wait for next update message and they iterate through entries in message. If entry has shortest path to destination, insert source as next hop to destination and record distance as distance from next hop to destination plus distance from this switch to next hop.

Question # 9

What is the concept of store and forward technology?

Answer

STORE AND FORWARD:

Data delivery from one computer to another is accomplished through store and forward technology. In this technology packet switch stores incoming packet and also forwards that packet to another switch or computer. For this purpose packet switch has internal memory into which it can hold packet if outgoing connection is busy. Packets for each connection held on queue.

Question # 10

How can Switched Virtual Network be established?

Answer

SWITCHED VIRTUAL CIRCUITS:

Most networks offer dynamic connections, which last for a relatively short time.

To handle this, ATM can dynamically establish a switched virtual circuit (SVC), allow it last as long as necessary and then terminate it.

The terminology comes from the Telco's where switching system normally refers to all switching.

ESTABLISHING AN SVC:

The computer sends a connection request to the switch to which it is attached.

Software in the switch finds a network path to the destination and sends along the connection request.

Each pair of switches in the path communicates to choose a VPI/VCI for their tables. Once the connection is established by the destination, a message is sent back to the originating computer to indicate the SVC is ready.

If any switch or the destination computer does not agree to setting up the VC, an error message is sent back and the SVC is not established

Question # 11

How can a bridge know whether to forward frames?

Answer

The bridge builds a list of MAC addresses on either side of the bridge. Therefore, it knows which packets should be forwarded to the other side and which should not. Most bridges are self-learning bridges. As soon as a frame arrives to a bridge, it extracts a source address from its header and automatically adds it in the list for that segment. In this way a bridge builds up address lists.

In the example of a packet that uses a MAC address not in its table it can err on the side of caution by forwarding the packet.

Question # 12

Compare connection oriented and connectionless Service.

Answer

Connection-Oriented vs. Connectionless Service

This characteristic specifies whether conversations take place in a more or less structured manner. When using a connection-oriented protocol, you incur the overhead of setting up a virtual circuit (a defined communications path) between the sender and receiver, which is maintained until the sender and receiver have completed their entire conversation.

When the conversation is completed, you incur the overhead of tearing down the virtual circuit. Connection-oriented protocols provide guaranteed delivery of messages in the order in which they were sent.

Contrast this with Connectionless service, which does not require establishing a session and a virtual circuit. This can be found in the network layer or transport layer, depending on the protocol. You can think of a connectionless protocol as being akin to mailing a post card. You send it and hope that the receiver gets it. Common features of a connectionless service are:

Packets do not need to arrive in a specific order

Reassembly of any packet broken into fragments during transmission must be in proper order

No time is used in creating a session

No Acknowledgement is required.

The largest connectionless network in use today is the Internet

Question # 13

Which type of information is obtained from network sniffer and in which mode Network sniffer operates?

Answer:

A network analyzer also called network monitor or a network sniffer is used to examine the performance of or debug a network. It can report statistics such as capacity utilization, distribution of frame size, collision rate or token circulation time

Most installations still use DIX Ethernet encoding in which there is no LLC/SNAP header in the frame. A network analyzer can tell from the values in the type field (small values are lengths, which mean an LLC/SNAP header is located in the first octets of the data area; large values are types, which mean no LLC/SNAP header is included).

The operation of network analyzer is a computer with a network interface that receives all frames, which is called promiscuous mode. So many desktop computers have interface that can be configured for promiscuous mode. When combined with software computer can examine any frame on LAN. In this way the communication across LAN is guaranteed to be private. Computer receives and displays frames on the LAN.

Network analyzer can be configured to filter and process frames. It can count frames of specific type or size. It displays only frames from or to specific computers.

Question # 14

Differentiate b/w Thin Ethernet, Thick Ethernet and Twisted Pair Ethernet?

Answer

Thick Ethernet:

Thick Ethernet, officially known as 10 Base-5, is the oldest form of Ethernet.

One form of cabling supported by Ethernet is low-loss 50 Ohm coaxial cable as shown in the figure below. This type of cable is 0.5" diameter (usually supplied with a yellow outer PVC coating) and rather inflexible. It has become known in the communications industry as "Thick Ethernet". The official name for this cable is 10 Baseband5 (10B5), indicating that it is specified for baseband communications at 10 Mbps over distances up to 500m.

Thin Ethernet:

Thin Ethernet, officially called 10 Base-2, is a less expensive version of 10 Base-5 (Thick Ethernet) technologies. It uses a lighter and thinner coaxial cable and dispenses with the external transceivers used with 10 Base-5.

10 Base-2 uses an RG-58A/U coaxial cable and is wired in a bus topology. Each device on the network is connected to the bus through a BNC "T" adapter, and each end of the bus must have a 50 Ohm terminator attached. Each node on the bus must be a minimum of 0.5 meters (1.5 feet) apart, and the overall length of the bus must be less than 185 meters (606 feet).

Twisted Pair Ethernet:

Twisted Pair Ethernet (10baseT), sometime also called "UTP" from "Unshielded Twisted Pair", is based on using a cable similar to phone-wiring. The cable is connected via RJ-45 connectors to the network card installed in the PC.

Question # 15

LIST SOME CHARACTERISTICS OF A CLIENT?

Answer

The characteristics of a client are explained below:

"Client is an arbitrary application program.

"It becomes client temporarily.

"It can also perform other computations.

"It is invoked directly by the user.

"It runs locally on the user's computer.

"It actively initiates contact with a server.

"It contacts one server at a time.

Question # 16

Explain TCP Segment Format?

TCP uses single format for all messages. TCP uses the term segment to refer to a message. Each message sent from TCP on one machine to TCP on another machine uses this format including data and acknowledgement.

Question # 17

What is IPv6 Address Notation?

Answer

128-bit addresses unwidely in dotted decimal; requires 16 numbers:

105.220.136.100.255.255.255.0.0.18.128.140.10.255.255

69DC: 8864: FFFF: FFFF: 0:1280:8C0A: FFFF

Zero-compression – series of zeroes indicated by two colons

FF0C: 0:0:0:0:0:0:B1

FF0C::B1

IPv6 address with 96 leading zeros is interpreted to hold an IPv4 address

Question # 18

How much data can present on original Ethernet segment at one time? To find out compute the delay-throughput product. The original Ethernet operate at 10 mega bits per second, and a segment was limited to 500 meters. Assume the signals propagate down the cable at 66 percent of the speed of light?

Answer

Speed of light = 299 792 458 m / s

Bandwidth=10

Delay = 66 percent of speed of light =197863022 m/s

Bandwidth Delay product= 247328777500

Question # 19

Explain these terms

Star topology, Ring topology, Frame?

Answer

FRAME:

A frame or hardware frame denotes a packet of a specific format on a specific hardware technology

RING TOPOLOGY

In this topology of network the computers are connected to each other in closed loop.

STAR TOPOLOGY

In this topology, all computers are attached to a central point, which is sometimes called the Hub

Question # 20

Explain the term bridge and its functions

Answer

A bridge keeps a list for each segment that consists of physical addresses of the computer attached to that segment. In this way a bridge knows on which segment a destination computer is attached.

Most bridges are self learning bridges. As soon as a frame arrives to a bridge, it extracts a source address from its header and automatically adds it in the list for that segment. In this way a bridge builds up address lists.

A bridge is a hardware device also used to connect two LAN segments to extend a LAN.

FRAME FILTERING

The most important task a bridge performs is frame filtering. If both the source and destination are on the same segment, it does not forward the frame to the other segment. A frame is forwarded to the other segment, if it is destined to that segment.

Question # 21

Explain NIC and Computer Processing?

Answer

NIC AND CPU PROCESSING:

NIC contains sufficient hardware to process data independent of system CPU. In which some NICs contain separate microprocessor. In addition to this it also include analog circuitry interface to system bus, buffering and processing

Question # 22

What is Parity Checking? Explain

Answer

Parity checking

To detect the error there are different schemes for which parity checking is also common used, in parity checking, parity refers to the number of bits set to 1 in the data item. A parity bit is an extra bit transmitted with data item chose to give the resulting bit even or odd parity

Question # 23

What do you know about Djikstra's algorithm?

Answer

Djikstra's algorithm

Djikstra's algorithm can accommodate weights on edges in graph. The shortest path is then the path with lowest total weight.

Question # 24

What are the uses of Repeater?

Answer

Repeater is a networking device

A repeater is used to increase the signal strength. It amplifies the weakening signal received from one segment and then retransmits onto another segment.

Qno.1 in LAN of 8 pcs, at low cost which wire scheme you will use. Give reasons

Answer:-

Reasons:-

5. Twisted pair Ethernet is the cheapest wiring that makes it so popular.
6. To allow changing the wiring without changing the interface hardware, NICs support
7. Multiple wiring schemes are not limited to Ethernet technology.
8. hubs (physically a star topology) to simulate a logical ring network.
9. The distance between devices
10. The amount of em-noise

REF:: Handouts Page No. 45,47

Qno.2 how many bits contains in VCI & VPI

Answer:-

4. Virtual Channel Indicator (16-bits)
5. Virtual Path Indicator (8- bit),

REF:: Handouts Page No. 67

Qno.3 we have a bridge network that has two segments. is that possible to transfer data simultaneously through segments ?

Unlike a repeater, a bridge uses two NICs to connect two segments. It listens to all traffic and recognizes frame format. It also forwards only correct complete frames and discards the collided and error frames. A typical bridge has two NICs, a CPU a memory and a ROM. It only runs the code stored in its ROM. A network bridge can be used to segment one network into two, thereby reducing the number of devices competing for transmission privileges.

For example,

if network A has 20 devices, there is the likelihood that two or more of them will attempt to transmit data at the same time and cause a collision. If a network bridge is added, it can split network A into networks A and B of 10 devices each.

REF:: Handouts Page No. 50

Qno.3 is there any technology that use LAN and WAN both topologies. if yes then write its Name

Star Topology, all computers are attached to a central point, which is sometimes called the Hub
Ring Topology network the computers are connected to each other in closed loop. In this network first computer passes data to the second and then second passes data to third and so on,

REF:: Handouts Page No. 25

Qno.4 why packet switch check the header frame before forwarding the packet

Answer:-

Data delivery from one computer to another is accomplished through store and forward technology. In this technology packet switch stores incoming packet and also forwards that packet to another switch or computer. For this purpose packet switch has internal memory into which it can hold packet if outgoing connection is busy. Packets for each connection held on queue.

[REF:: Handouts Page No. 56](#)

Qno.5 Why the designer of ATM technology select cells not packets? Give five reasons

Answer:-

Reason:

6. ATM multiplexed these 53-byte cells instead of packets which reduced worst-case cell contention jitter by a factor of almost 30, reducing the need for echo cancellers.
7. Reduction of jitter (and also end-to-end round-trip delays)
8. Full-length 1500 byte (12000-bit) data packet would take 77.42 to transmit.
9. The design of ATM aimed for a low-jitter network interface
10. ATM broke up all packets, data, and voice streams into 48-byte chunks, adding a 5-byte routing header to each one so that they could be reassembled later

[REF:: Handouts Page No. 66](#)

Qno.6 If two far away situated offices are to connect with will be appropriate LAN extension or WAN. Give four reasons to support you answer

8. If two buildings are located far from each other, a bridge, a pair of fiber modems and an optical fiber can be used to connect two LANs
9. LAN segments that need to be interconnected but are slightly far away, point to point wireless solutions are available
10. They are generally kept over a tower so that line of sight is realized between the locations, for best performance.
11. These outdoor access points are configured to operate in the point-to-point or point-to-multipoint modes and one such access point along with antenna/tower is required in each department/building which needs to be interconnected.
12. In certain areas, metro Ethernet might be available where the service provider has the infrastructure to connect two locations of a same branch located in different places at speeds of 10/100 Mbps using his public network infrastructure

Qno.7 writes three benefits of routing if source address is not included in the header

Answer:-

Benefit

It allows fast and efficient routing.

Packet switch need not have complete information about all destinations in spite it just has next The network can continue to function even if topology changes without notifying entire network

[REF:: Handouts Page No. 58](#)

Qno.8 How many methods are available to span a network over longer distances? Just write their names. (2 Marks)

Answer:-

- LEASED SERIAL LINE CONNECTION:
- LEASED SATELLITE CHANNEL:

REF:: Handouts Page No. 52

Qno.9 which two identifier in ATM are used to make Virtual Circuit? (2 Marks)

Answer:-

VCI (Virtual Channel Indicator)

VPI (Virtual Path Indicator)

REF:: Handouts Page No. 67

Qno.10 conceptual question about Virtual private Network.(marks 3)

Answer:-

Virtual Private Network (VPN) combines the features of both private and public networks. It is limited to single organization and uses public network for connectivity. These connections are sometimes called tunnels and connect sites. Each site sees tunnel as point-to-point link. There is no access for other users of public networks.

REF:: Handouts Page No. 75

Qno.11 Describe the procedures in steps of Telephone calling, A connection oriented service (Marks5)

Answer:

The connection-oriented service paradigm for networking is similar to the manner in which telephones are used. This is given as follows:

A caller dials a number of the destination. The telephone at the destination signals the arrival of a connection request. If the called person does not answer; the caller gives up after waiting for a timeout. If the called person does answer, then the connection is established as binary connection identifier is given to each of the two parties to enable identification of the connection.

REF:: Handouts Page No. 67

Qno.12 Describe two major components of thick Ethernet (2)

Answer:-

- ❑ Network interface card (NIC)
- ❑ Analog electronic device

REF:: Handouts Page No. 42

Qno.13 Suppose in a computer network of a chemical company, each packet is transferred with equal amount of time between systems. Consider this scenario what will be the jitter?(2)

Answer:-

The amount of delay that a network introduces is called jitter. A network with zero jitter takes exactly the same amount of time to transfer each packet. A network with high jitter takes much longer to deliver some packets than others

REF:: Handouts Page No. 80

Qno.14 Suppose in university campus, bridges are used in its local are network. Explain all steps, how bridge build up address list table of all attached computers. (3)

Answer:-

University campus implement bridges to attached the computers, bridges as a frame arrives to it, extracts a source address from its header and automatically adds it in the list for that segment. In this way a bridge builds up address lists.

Event	Segment 1 List	Segment 2 List
Bridge boots	—	—
U sends to V	U	—
V sends to U	U, V	—
Z broadcasts	U, V	Z
Y sends to V	U, V	Z, Y
Y sends to X	U, V	Z, Y
X sends to W	U, V	Z, Y, X
W sends to Z	U, V, W	Z, Y, X

REF:: Handouts Page No. 50

Qno.15 Suppose an organization wants voice communication in a network by using ATM technology. Explain how ATM eliminates the varying delays associated with variable size data packets.

Answer:-

Organization wants voice communication introduced ATM to meet several goals. It provides universal service for all subscribers and support for all users for voice, video and data. It has a single unified infrastructure (no separate LANs and WANs). It gives guaranteed service when it is appropriate and support for low cost devices.

JITTER:

Jitter is the term used for variance in transmission delays. Jitter is significance for voice, video and data. In LANs, jitter can occur when a packet is delayed because the network is busy.

PACKET SIZES:

Large packets result in fewer overheads because a smaller fraction of the packet is used for header information. Optimum networks use 4kB packets or larger. Large packets can't easily be used for voice for example 8-bit samples (at 125usec per sample) would require half a second to fill a 4kB packet. Echo cancellation can only be used with low transmission delays.

REF:: Handouts Page No. 66

Qno.16 While designing the ATM technology why designer of ATM chosen cells over packets which being used widely? Give five reasons to support your answers.(5)

Reason:

- ATM multiplexed these 53-byte cells instead of packets which reduced worst-case cell contention jitter by a factor of almost 30, reducing the need for echo cancellers.
- Reduction of jitter (and also end-to-end round-trip delays)
- Full-length 1500 byte (12000-bit) data packet would take 77.42 to transmit.
- The design of ATM aimed for a low-jitter network interface

5. ATM broke up all packets, data, and voice streams into 48-byte chunks, adding a 5-byte routing header to each one so that they could be reassembled later

Qno.17 Suppose star organization using repeater in its network. If frames comes from another network then will the repeater be capable of distinguish between frames from two different networks? Support your answer with a reason also writes down three drawbacks, this organization may face due to use of repeater.

Answer:-

Drawbacks

24. Repeaters provide no method for isolating traffic generated on one cable segment from traffic generated by the other cable segment
25. When network uses a repeater to connect cable segment A to segment B whether or not there is a station in segment B that is the destination of the signal
26. Ethernet standard specifies that any two stations cannot be separated by more than four repeaters.
27. Repeaters do not recognize frame formats; they just amplify and retransmit the electrical signal. If a collision or error occurs in one segment, repeaters amplify and retransmit also the error onto the other segments.

Question:

What is Broadcast?

Answer:

Broadcast means to sent to more than one recipient. In communications and on networks, a broadcast message is one distributed to all stations. For example, as in television or

radio networks, there is one transmitter and many listeners

Question:

I can't understand it. " Instead of how a specific protocol handles congestion, we should concentrate on what congestion is and why it must be handled."

Answer:

It is very simple english and it means that for now we do not need to go into details that how the congestion (overloading) is handles, but we are more to focus on what is congestion and why it is needed to be controlled.

Question:

I can't understand the basic concept of "Fixed networks". Plz explain the term individually and also with respect to the mobile networks

Answer:

Fixed Network includes the example of your lab network with laid network lines and are not moved usually. On the other side mobile networks include example of your Mobile phones which can also be said as wireless networks.

Question:

“Multiple technologies exist that are used to connect different networks” So my question is what are these different technologies? Kindly list them and also list the features of these technologies and why we don't have any single standard for connecting networks?.

Answer:

You do not need to go into details of the multiple networks provided by different companies.

Some of the examples of the networking technologies are: Ethernet, Fast Ethernet, Myrinet,

ATM, SCRAMNet, Fibre Channel, FDDI. If you need to learn about the features of these technologies for your own knowledge, you should search for them over the internet as a vast

information exists about these technologies over the internet.

Different companies develop their own technology for their sale and publicity to compete with

other companies as well as they provide with some enhance idea with the existing technology

that is then considered as more improved version or given different name.

Question:

Explain about routers and how different networks are connected by routers?

Answer:

Router is an agent device on a communications network that speeds up message delivery. On a single network linking many computers through a mesh of possible connections, a

router receives transmitted messages and forwards them to their correct destinations over the

most efficient available route. On an interconnected set of local area networks (LANs) using the

same communications protocols, a router serves the somewhat different function of acting as a

link between LANs, enabling messages to be sent from one to another. You will read in detail

about routers in later lectures.

Question:

Explain the term Latency.

Answer:

Latency is the time required for a signal to travel from one point on a network to another.

Question:

How models are either so simple are either so complex how can we explain by giving any example?

Answer:

These conceptual models complexity or simplicity depends on the information provided on the provider side or the understanding gathered on the learner (client) side. It can be

easily understood with your own example. As it is written in very clear words in handouts that

the complexity depends if the information provided is not prior to details or they are so simple

that sometimes it is hard to distinguish between the details. But you have not understood it, so

you have taken it as the complex form and don't find it that the information provided is enough

to your understanding.

Question:

I want you to explain the concept of PING command.

Answer:

PING stands for Packet Internet Groper. It is a protocol for testing whether a particular computer is connected to the Internet by sending a packet to its IP address and waiting

for a response. In any windows system , 95, 98 , XP or 2000 or any other, go to the windows command prompt and type "ping/?", press enter and you will know if it is working in the windows and also how can you use different options of PING command.

Question:

How can we disable a pinging request (for security purpose)?

Answer:

PING is an internal command which cannot be disabled. Also , there is no security issue in it that it is needed to be disabled. It only provies the networking information and IP information which is open to the clients.

Question:

'tracert' does not work in the LAN when sitting on a client computer, what could be the possible causes and how can be resolved?

Answer:

Trace Route works on all the systems on the network. You should see if you are not making any typing mistake or if you are using Wrindows XP then you need to type "tracert" command for trace routing.

Question:

PING command is only used to know the existence of a site or is there any other use/advantage too of PING command.

Answer:

It is a protocol for testing whether a particular computer is connected to the Internet/LAN by sending a packet to its IP address and waiting for a response.

Question:

What factor determines how much byte data packed is send from the host computer. In one of the example in Lecture 2 you determines 32 byte and in other it was 56?

What this difference means?

Answer:

It is an internal feature of the ping command that it by default work with 32 data bytes. You simply try the ping command and you will learn its working , ping with different

servers. In any windows system , 95, 98 , XP or 2000 or any other, go to the windows command

prompt and type "ping/?", press enter and you will know if it is working in the windows and also how can you use different options of PING command.

Question:

What is the difference b/w "PING and Trace Route Probing Tools".

Answer:

PING stands for Packet Internet Groper. It is a protocol for testing whether a particular computer is connected to the Internet by sending a packet to its IP address and waiting

for a response. In any windows system , 95, 98 , XP or 2000 or any other, go to the windows command prompt and type "ping/?", press enter and you will know if it is working in the windows and also how can you use different options of PING command.

Traceroute shows all the computers in the defined network path, it shows all the computers which comes in the way of source and the destination computers. Type "tracert/?" in command prompt in Windown XP and you will see its different functionalities.

www.virtualians.pk Prepared by: Irfan Khan

Question:

As ping tells us about the connectivity of the host or other computer, is it uses any resources of the local computer or not?

Answer:

Yes, it does use resources of the computer. It is an internal command and use the network resources.

Question:

What is the difference between computational power and resources?

Answer:

Computational power can be defined as the act of processing of the computer.

Resource is any part of a computer system or a network, such as a disk drive, printer, or memory,

that can be allotted to a program or a process while it is running.

Question:

I tried ping and tracert both with different sites but when I do it with VU website it does not respond and messages come out to be request time out. Why is it so? As VU website is alive, is that because of any security reason or there is some other problem?

Answer:

It is not any security reason. Any site which is outside the secure link responds to ping and tracert commands. The problem is the network traffic over the VU sites or the internet

speed you are having. As VU sites holds a lots of users and these commands need a free way for

information. I can tell you as an example, if i use ping or tracert commands for vu sites they

work very correctly and efficiently without any time out, but while using any other site, such as

google.com or msn.com, i have tried to use tracert command on both sites with an interval of half

an hour, and i have come up with the result that when i used the command earlier it was working

ok and just gave a timeout once , but the other time it is not responding at all.

Question:

Why Some networks or computers reject the ping packets ?

Answer:

It may be due to the complexity of the network connections. Also it can be due to the fact that the server is too busy and not been able to respond all the clients. This may happen

mostly in internet probing.

Question:

Is this ping & tacing a route tools only use in XP window or other windows like 98 or 95 etc.?

Answer:

In any windows system , 95, 98 , XP or 2000 or any other, go to the windows command prompt and type "ping/?", press enter and you will know if it is working in the windows and also how can you use different options of PING command.

Question:

Kindly explain briefly the basics of Data Stuffing?

Answer:

Data Stuffing basically involves bit stuffing and byte stuffing: Bit stuffing is the practice of adding bits to a stream of data. Bit stuffing is required by many network and communications protocols for the following reasons: To prevent data being interpreted as control

information. For protocols that require a fixed-size frame, bits are sometimes inserted to make

the frame size equal to this set size.

For protocols that required a continuous stream of data, zero bits are sometimes inserted to

ensure that the stream is not broken.

Byte Stuffing , also referred to as Character Stuffing.

Question:

What is CSMA ? What is the concept of LAN hardware?

Answer:

CSMA is short for Carrier Sense Multiple Access / Collision Detection, a set of rules determining how network devices respond when two devices attempt to use a data channel

simultaneously (called a collision). Standard Ethernet networks use CSMA/CD to physically

monitor the traffic on the line at participating stations. If no transmission is taking place at the

time, the particular station can transmit. If two stations attempt to transmit simultaneously, this

causes a collision, which is detected by all participating stations. After a random time interval,

the stations that collided attempt to transmit again. If another collision occurs, the time intervals from which the random waiting time is selected are increased step by step. This is known as

exponential back off.

CSMA/CD is a type of contention protocol. Networks using the CSMA/CD procedure are simple

to implement but do not have deterministic transmission characteristics.

LAN hardware includes the network adapter and the wires used to connect these adaptors in all

computers. This is the simplest definition of LAN hardware. What do you not understand about

it, specify your confusion.

Question:

What is the basic function of trace routing and ping.?

Answer:

PING stands for Packet Internet Groper. It is a protocol for testing whether a

particular computer is connected to the Internet by sending a packet to its IP address and waiting

for a response. In any windows system , 95, 98 , XP or 2000 or any other, go to the windows command prompt and type "ping/?", press enter and you will know if it is working in the windows and also how can you use different options of PING command.

Traceroute shows all the computers in the defined network path, it shows all the computers

which comes in the way of source and the destination computers. Type "tracert/?" in command prompt in Windows XP and you will see its different functionalities.

Question:

Please briefly explain "binary exponential backoff".

Answer:

If two stations attempt to transmit simultaneously, this causes a collision, which is detected by all participating stations. After a random time interval, the stations that collided attempt to transmit again. If another collision occurs, the time intervals from which the random waiting time is selected are increased step by step. This is known as exponential back off.

Question:

Broadcast Address must be "all 1s" Pls tell me that what are the total no. of 1s in a broadcast address. and how it can be sent to other computers on the LAN?

Answer:

A broadcast message is addressed to all stations on the network. The destination address in a broadcast message consists of all 1s (0xFFFFFFFF). All stations automatically receive frames with this address. Normally, broadcast messages are sent for network management and diagnostic purposes.

On IP networks, the IP address 255.255.255.255 (in binary, all 1s) is the general broadcast address. You can't use this address to broadcast a message to every user on the Internet because

routers block it, so all you end up doing is broadcasting it to all hosts on your own network. The

broadcast address for a specific network includes all 1s in the host portion of the IP address. For

example, on the class C network 192.168.1.0, the last byte indicates the host address (a 0 in this

position doesn't refer to any host, but provides a way to refer to the entire network). The value

255 in this position fills it with all 1s, which indicates the network broadcast address, so packets

sent to 192.168.1.255 are sent to all hosts on the network. So the broadcast address contains 8

number of 1's.

Question:

In CSMA/CD why we use a random time from 0-d. Why we not assign a specific but different wait time for all the machines.

Answer:

Standard Ethernet networks use CSMA/CD to physically monitor the traffic on the line at participating stations. If no transmission is taking place at the time, the particular

station can transmit. If two stations attempt to transmit simultaneously, this causes a collision,

which is detected by all participating stations. After a random time interval, the stations that

collided attempt to transmit again. If another collision occurs, the time intervals from which the

random waiting time is selected are increased step by step. That is why we use random time.

Question:

Is CRC detects balance errors or not. If not then is there any technique that can

detect balance error ?

Answer:

CRC can detect balance errors. CRC error checking uses a complex calculation to generate a number based on the data transmitted which is same as balance errors.

Question:

What does Spatial and Temporal Locality Principles mean? Does it mean that we are having LANs for in LANS computer are located in a close contact with eath other, and to

eastablish a connection with nearby computer is a lot more easier than with the computers residing far away?

Answer:

Spatial Locality pattern refers to the communication which is carried out between computers in a network which are located very near to each other.

Temporal Locality pattern means that the communicaion is carried out between same computers

again and again.

It may cause difference in peer to peer connection speed, but when it comes to LANs with servers then the speed is controlled by the server.

Question:

How can perform shift register?

Answer:

Shift registers are a type of sequential logic circuit, mainly for storage of digital data. They are a group of flip-flops connected in a chain so that the output from one flip-flop

becomes the input of the next flip-flop. Most of the registers possess no characteristic internal

sequence of states. All the flip-flops are driven by a common clock, and all are set or reset simultaneously.

Revise your CS302 course to remember Shift Registers.

Question:

What is IPV4 and IPV6?

What is the difference between "ping" and "ping6" and "tracert" and "tracert6"?

Answer:

The core protocol used on the Internet is called the Internet Protocol or simply IP.

The current version of this protocol is version 4 (IPv4). This version of the protocol was designed over 20 years ago. Since that time, a great deal about networking has been learned and

much more still has changed. The original design has proven to work well, but in hindsight, a

number of shortcomings with the design have been exposed. A new version of the protocol, version 6, is being deployed by the Internet community (version 5 was used for testing and never

widely implemented). Most systems now have IPv6 support built in and many network operators

are beginning trials and building production networks based on the newer version of IP.

IPv4 Internet addresses are unique 32-bit identifiers used to address hosts on the Internet.

IPv6 is backward compatible with IPv4, provided that special techniques are used. For example,

to enable communication between far distances of IPv6 devices connected by IPv4 networks,

tunneling may be employed. To support IPv4/IPv6 compatibility, a scheme was developed to

allow IPv4 addresses to be embedded within the IPv6 address structure. This method takes regular IPv4 addresses and puts them in a special IPv6 format so they are recognized as being

IPv4 addresses by certain IPv6 devices.

ping6 is the IPv6 version ping.

To trace a full IPv6 network path, you can use the tracert6 command.

Question:

What are the pros and cons of TCP /IP network?

Answer:

TCP/IP (Transmission Control Protocol/Internet Protocol) allows computers to communicate with one another over a network. Some of the advantages of TCP/IP include:

- Computers on the same network can communicate
- Computers on different networks can communicate
- Different types of computers can send messages to one another.
- Platform independent
- The protocol of the internet

TCP/IP is really a standard which computer manufacturers have agreed will be used to allow

computers to exchange data. TCP/IP is not just one protocol, but is actually a collection of several. TCP and IP are the two major components, although there are several other components

included in the entire procedure.

Question:

Please explain the term " TDM" . What is it ?

Answer:

TDM stands for Time Division Multiplexing. It is a type of multiplexing where

two or more channels of information are transmitted over the same link by allocating a different

time interval ("slot" or "slice") for the transmission of each channel. I.e. the channels take turns to use the link. Some kind of periodic synchronising signal or distinguishing identifier is usually

required so that the receiver can tell which channel is which. TDM becomes inefficient when

traffic is intermittent because the time slot is still allocated even when the channel has no data to transmit.

Question:

As the concept of packets if we want to send a big file in Mb's so we have to

break it into packets as first turn into Kb's then we can send the file if we want to send it quickly?

Answer:

If for example you are required to send a 5MB file over a network. If you send

the whole file as one file, it will first break into packets and then will be sent and these packets

will reunite to for the original file. If you break it into file of size 1MB each or 500KB each then

it will again be broken down in packets and same procedure will be carried out. But the packets

in case of whole 5MB file, there will be many packets and so it will take a little more time while

creating and reuniting packets as compared to the less number of packets in case of 1MB or

500KB files, which will take a little less time.

Question:

Computer Networks or individual servers are connect to form THE INTERNET

via Routers. When we use "trace route" command to find the path from source to destination, are these IP's belong to routers, or servers or both because router must also have some address to be

uniquely identified. If both routers and server machines have some IP address then how could we

distinguish between these IP's?

Answer:

Routers do not have any IP address. They are the electronic devices. Routers are an intermediary device on a communications network that expedites message delivery. On a

single network linking many computers through a mesh of possible connections, a router receives transmitted messages and forwards them to their correct destinations over the most

efficient available route. On an interconnected set of local area networks (LANs) using the same

communications protocols, a router serves the somewhat different function of acting as a link

between LANs, enabling messages to be sent from one to another. Therefore you only need one

IP address. i.e., of server.

Question:

What is the difference between Transponder and repeaters?

Answer:

Transponders are devices that receives a signal from and retransmits it on a different frequency to one or more other destinations.

Repeaters are devices used on communications circuits that decreases distortion by amplifying or

regenerating a signal so that it can be transmitted onward in its original strength and form. On a

network, a repeater connects two networks or two network segments at the physical layer and

regenerates the signal.

Question:

Plz explain what is difference between character based machines and bit oriented machines?

Answer:

Bit Oriented Machines are based on Bit Stiffing Techniques.

Bit stuffing is the practice of adding bits to a stream of data. Bit stuffing is required by many

network and communications protocols for the following reasons:

To prevent data being interpreted as control information.

For protocols that require a fixed-size frame, bits are sometimes inserted to make the frame size

equal to this set size.

For protocols that required a continuous stream of data, zero bits are sometimes inserted to

ensure that the stream is not broken.

Character Oriented Machines are based on Byte Stiffing techniques. See handouts for byte stuffing detials.

Question:

What is the diffrence between LAN user IP and a Modem user IP ? Most of the

LAN IP's of diffrent LANs are same for example 192.168.0.43 many diffrent LAN users have

this IP so how can i acess this IP of diffrent LAN network user which have the same IP ? Is this

poossible the modem IP are also same of diffrent users ?

Answer:

LAN IP is the IP address given to a network (to every individual computer). This

IP address which you have mentioned is the standard of any LAN IP network address. Modem IP

is is the IP given at the time when you are connected to the ISP (Internet Service Provider) which

gives a different IP to all its users when they are connected. IP is basically managed by the servers. In a LAN configuration, each computer has a different IP address. It may be so if you

visit two different labs, you may find same IP address on some computer in these two labs, but

these labs will not be interconnected to each other. Similarly the modem IP's are never the same.

In simple words IP addresses are not same on similar networks.

Question:

I want to know what is parity checking ? When we will CRC ?

Answer:

Parity Checking is the procedure of using parity bit to check the accuracy of transmitted data.

CRC (cyclic redundancy check) is a procedure used in checking for errors in data transmission.

CRC error checking uses a complex calculation to generate a number based on the data transmitted. The sending device performs the calculation before transmission and sends its result

to the receiving device. The receiving device repeats the same calculation after transmission. If

both devices obtain the same result, it is assumed that the transmission was error-free. The procedure is known as a redundancy check because each transmission includes not only data but

extra (redundant) error-checking values.

Question:

Please explain the two terms "Byte Stuffing" & "Bit Stuffing". What is the difference between Byte Stuffing and Bit Stuffing ?

Answer:

Bit stuffing is the practice of adding bits to a stream of data. Bit stuffing is required by many network and communications protocols for the following reasons:

To prevent data being interpreted as control information.

For protocols that require a fixed-size frame, bits are sometimes inserted to make the frame size

equal to this set size.

For protocols that required a continuous stream of data, zero bits are sometimes inserted to

ensure that the stream is not broken.

Byte Stuffing , also referred to as Character Stuffing.

Question:

What is meant by impairment?

Answer:

Impairment means to cause a damage or lost of some network connection.

Question:

How CRC hardware actually work with data ?

Answer:

CRC (cyclical redundancy check) is a procedure used in checking for errors in

data transmission. CRC error checking uses a complex calculation to generate a number based on

the data transmitted. The sending device performs the calculation before transmission and sends

its result to the receiving device. The receiving device repeats the same calculation after

transmission. If both devices obtain the same result, it is assumed that the transmission was

error-free. The procedure is known as a redundancy check because each transmission includes

not only data but extra (redundant) error-checking values.

Question:

Which is the better option for communicating (point to point) like star topology or the other one?

Answer:

It depends on the network you are using. All topologies have their advantages and disadvantages depending on the usage. Star topology is mostly used in general networks.

Question:

What is the meaning of Mobile Network and Semi Persistent connections ?

Answer:

Mobile Networks are a kind of wireless networking in which there is no physical or direct link through wires as in your lab network. Example of mobile networks are the mobile phones.

Semi persistent connections means those connections which are with intervals. For example, you

might have noticed that sometimes your mobile phones just reply to some network even when you

are not receiving or dialing any number, it is because the network keeps on checking the devices

in range not all the time but after several intervals.

Question:

What is firewall restriction from ISP?

Answer:

A firewall is installed to prevent computers in the network from communicating directly with computers external to the network and vice versa. Instead, all communication is

routed through a proxy server outside of the network, and the proxy server decides whether it is

safe to let a particular message or file pass through to that network. ISPs usually restrict the

protocol which allows the IP address to travel along the path to keep safe from any kind of access to their network not allowing to travel through their proxy.

Question:

What is hardware fault? where it occur and why?

Answer:

Hardware fault can occur if the system crashes, its physical components cause errors due to any reason like system halt, over heating, component failure damage. There can be

many reasons of its occurrence. Please contact your operating systems instructor for more details.

Question:

What are the two types of transmission technology available?

Answer:

(i) Broadcast and (ii) point-to-point

Question:

What is subnet?

Answer:

A generic term for section of a large networks usually separated by a bridge or router.

Question:

Difference between the communication and transmission.

Answer:

Transmission is a physical movement of information and concern issues like bit polarity, synchronisation, clock etc. Communication means the meaning full exchange of information between two communication media.

Question:

What are the possible ways of data exchange?

Answer:

(i) Simplex (ii) Half-duplex (iii) Full-duplex.

Question:

What is SAP?

Answer:

Series of interface points that allow other computers to communicate with the other layers of network protocol stack.

Question:

What is NETBIOS and NETBEUI?

Answer:

NETBIOS is a programming interface that allows I/O requests to be sent to and received from a remote computer and it hides the networking hardware from applications.

NETBEUI is NetBIOS extended user interface. A transport protocol designed by microsoft and

IBM for the use on small subnets.

Question:

What is RAID?

Answer:

A method for providing fault tolerance by using multiple hard disk drives.

Question:

What is passive topology?

Answer:

When the computers on the network simply listen and receive the signal, they are referred to as passive because they don't amplify the signal in any way. Example for passive

topology - linear bus.

Question:

What is Brouter?

Answer:

Hybrid devices that combine the features of both bridges and routers.

Question:

What is cladding?

Answer:

A layer of a glass surrounding the center fiber of glass inside a fiber-optic cable.

Question:

How Gateway is different from Routers?

Answer:

A gateway operates at the upper levels of the OSI model and translates information between two completely different network architectures or data formats

Question:

What is attenuation?

Answer:

The degeneration of a signal over distance on a network cable is called attenuation.

Question:

What is MAC address?

Answer:

The address for a device as it is identified at the Media Access Control (MAC) layer in the network architecture. MAC address is usually stored in ROM on the network adapter card and is unique.

Question:

Difference between bit rate and baud rate.

Answer:

Bit rate is the number of bits transmitted during one second whereas baud rate refers to the number of signal units per second that are required to represent those bits. baud rate

= bit rate / N where N is no-of-bits represented by each signal shift

Question:

What are the types of Transmission media?

Answer:

Signals are usually transmitted over some transmission media that are broadly classified in to two categories.

Question:

Guided Media:

Answer:

These are those that provide a conduit from one device to another that include

twisted-pair, coaxial cable and fiber-optic cable. A signal traveling along any of these media is

directed and is contained by the physical limits of the medium. Twisted-pair and coaxial cable

use metallic that accept and transport signals in the form of electrical current. Optical fiber is a

glass or plastic cable that accepts and transports signals in the form of light.

Question:

Unguided Media:

Answer:

This is the wireless media that transport electromagnetic waves without using a physical conductor. Signals are broadcast either through air. This is done through radio communication, satellite communication and cellular telephony.

Question:

What is Project 802?

Answer:

It is a project started by IEEE to set standards to enable intercommunication between equipment from a variety of manufacturers. It is a way for specifying functions of the physical layer, the data link layer and to some extent the network layer to allow for interconnectivity of major LAN protocols. It consists of the following: 802.1 is an internetworking standard for compatibility of different LANs and MANs across protocols. 802.2

Logical link control (LLC) is the upper sublayer of the data link layer which is non-architecture-

specific, that is remains the same for all IEEE-defined LANs. Media access control (MAC) is the

lower sublayer of the data link layer that contains some distinct modules each carrying

proprietary information specific to the LAN product being used. The modules are Ethernet LAN

(802.3), Token ring LAN (802.4), Token bus LAN (802.5). 802.6 is distributed queue dual bus

(DQDB) designed to be used in MANs.

Question:

What are the different type of networking / internetworking devices?

Answer:

Repeater: Also called a regenerator, it is an electronic device that operates only at physical layer. It receives the signal in the network before it becomes weak, regenerates the original bit pattern and puts the refreshed copy back in to the link. **Bridges:** These operate both in

the physical and data link layers of LANs of same type. They divide a larger network in to smaller segments. They contain logic that allow them to keep the traffic for each segment separate and thus are repeaters that relay a frame only the side of the segment containing the

intended recipient and control congestion. **Routers:** They relay packets among multiple interconnected networks (i.e. LANs of different type). They operate in the physical, data link and

network layers. They contain software that enable them to determine which of the several

possible paths is the best for a particular transmission. **Gateways:** They relay packets among

networks that have different protocols (e.g. between a LAN and a WAN). They accept a packet

formatted for one protocol and convert it to a packet formatted for another protocol before forwarding it. They operate in all seven layers of the OSI model.

Question:

What are the data units at different layers of the TCP / IP protocol suite?

Answer:

The data unit created at the application layer is called a message, at the transport

layer the data unit created is called either a segment or an user datagram, at the network layer the

data unit created is called the datagram, at the data link layer the datagram is encapsulated in to a frame and finally transmitted as signals along the transmission media.

Question:

What is difference between ARP and RARP?

Answer:

The address resolution protocol (ARP) is used to associate the 32 bit IP address

with the 48 bit physical address, used by a host or a router to find the physical address of another

host on its network by sending a ARP query packet that includes the IP address of the receiver.

The reverse address resolution protocol (RARP) allows a host to discover its Internet address

when it knows only its physical address.

Question:

What is the minimum and maximum length of the header in the TCP segment and IP datagram?

Answer:

The header should have a minimum length of 20 bytes and can have a maximum length of 60 bytes.

Question:

What is the range of addresses in the classes of internet addresses?

Answer:

Class A 0.0.0.0 - 127.255.255.255 Class B 128.0.0.0 - 191.255.255.255 Class C

192.0.0.0 - 223.255.255.255 Class D 224.0.0.0 - 239.255.255.255 Class E 240.0.0.0 -

247.255.255.255

Question:

What are major types of networks and explain?

Answer:

Peer-to-peer network, computers can act as both servers sharing resources and as clients using the resources. Server-based networks provide centralized control of network resources and rely on server computers to provide security and network administration

Question:

What are the important topologies for networks?

Answer:

BUS topology: In this each computer is directly connected to primary network cable in a single line. Advantages: Inexpensive, easy to install, simple to understand, easy to extend. STAR topology: In this all computers are connected using a central hub. Advantages:

Can be inexpensive, easy to install and reconfigure and easy to trouble shoot physical problems.

RING topology: In this all computers are connected in loop. Advantages: All computers have

equal access to network media, installation can be simple, and signal does not degrade as much

as in other topologies because each computer regenerates it.

Question:

What is mesh network?

Answer:

A network in which there are multiple network links between computers to provide multiple paths for data to travel.

Question:

What is difference between baseband and broadband transmission?

Answer:

In a baseband transmission, the entire bandwidth of the cable is consumed by a single signal. In broadband transmission, signals are sent on multiple frequencies, allowing multiple signals to be sent simultaneously.

Question:

Explain 5-4-3 rule?

Answer:

In a Ethernet network, between any two points on the network ,there can be no more than five network segments or four repeaters, and of those five segments only three of segments can be populated.

Question:

What is the difference between routable and non- routable protocols?

Answer:

Routable protocols can work with a router and can be used to build large networks. Non-Routable protocols are designed to work on small, local networks and cannot be used with a router

Question:

Why should you care about the OSI Reference Model?

Answer:

It provides a framework for discussing network operations and design.

Question:

What is logical link control?

Answer:

One of two sublayers of the data link layer of OSI reference model, as defined by the IEEE 802 standard. This sublayer is responsible for maintaining the link between computers when they are sending data across the physical network connection.

Question:

What is virtual channel?

Answer:

Virtual channel is normally a connection from one source to one destination, although multicast connections are also permitted. The other name for virtual channel is virtual circuit.

Question:

What is traffic shaping?

Answer:

One of the main causes of congestion is that traffic is often bursty. If hosts could be made to transmit at a uniform rate, congestion would be less common. Another open loop method to help manage congestion is forcing the packet to be transmitted at a more predictable rate. This is called traffic shaping.

Question:

What is multicast routing?

Answer:

Sending a message to a group is called multicasting, and its routing algorithm is called multicast routing.

Question:

What is region?

Answer:

When hierarchical routing is used, the routers are divided into what we will call regions, with each router knowing all the details about how to route packets to destinations within its own region, but knowing nothing about the internal structure of other regions.

Question:

What is silly window syndrome?

Answer:

It is a problem that can ruin TCP performance. This problem occurs when data are passed to the sending TCP entity in large blocks, but an interactive application on the receiving side reads 1 byte at a time.

Question:

What is Mail Gateway?

Answer:

It is a system that performs a protocol translation between different electronic mail delivery protocols.

Question:

What is IGP (Interior Gateway Protocol)?

Answer:

It is any routing protocol used within an autonomous system.

Question:

What is EGP (Exterior Gateway Protocol)?

Answer:

It is the protocol the routers in neighboring autonomous systems use to identify the set of networks that can be reached within or via each autonomous system.

Question:

What is autonomous system?

Answer:

It is a collection of routers under the control of a single administrative authority and that uses a common Interior Gateway Protocol.

Question:

What is BGP (Border Gateway Protocol)?

Answer:

It is a protocol used to advertise the set of networks that can be reached within an autonomous system. BGP enables this information to be shared with the autonomous system.

This is newer than EGP (Exterior Gateway Protocol).

Question:

What is Gateway-to-Gateway protocol?

Answer:

It is a protocol formerly used to exchange routing information between Internet core routers.

Question:

What is NVT (Network Virtual Terminal)?

Answer:

It is a set of rules defining a very simple virtual terminal interaction. The NVT is used in the start of a Telnet session.

Question:

What is a Multi-homed Host?

Answer:

It is a host that has a multiple network interfaces and that requires multiple IP addresses is called as a Multi-homed Host.

Question:

What is OSPF?

Answer:

It is an Internet routing protocol that scales well, can route traffic along multiple paths, and uses knowledge of an Internet's topology to make accurate routing decisions.

Question:

What is Proxy ARP?

Answer:

It is using a router to answer ARP requests. This will be done when the originating host believes that a destination is local, when in fact it lies beyond router.

Question:

What is RIP (Routing Information Protocol)?

Answer:

It is a simple protocol used to exchange information between the routers.

Question:

LAN, MAN, CAN, and WAN: What are the differences? What is each one used for?

Answer:

LAN, MAN, CAN, and WAN are all different types of networks used when connecting to the internet and/or other computers. LAN is a local area network used in small areas such as homes or offices. LANs offer quick data transferring due to its small area and transfer area. Additionally, LANs don't need an external telecom device in order to be accessed.

MANs are metropolitan area networks, so they are used for larger areas, such as entire cities.

CANs are a type of MAN meaning campus area network. As the name implies, they are commonly used on academic campuses, but can also be used in most any moderate area. WAN is

a wide area network and is good for a larger area. WAN are often compared to LAN connections. They have a slower data transfer due to the wide range. WANs are commonly used in large organizations due to their extensive area.

Question:

What is a network firewall?

Answer:

A firewall is a system or group of systems that enforces an access control policy

between two or more networks. The actual means by which this is accomplished varies widely,

but in principle, the firewall can be thought of as a pair of mechanisms: one which exists to block

traffic, and the other which exists to permit traffic. Some firewalls place a greater emphasis on

blocking traffic, while others emphasize permitting traffic. Probably the most important thing to

recognize about a firewall is that it implements an access control policy. If you don't have a good

idea of what kind of access you want to allow or to deny, a firewall really won't help you. It's

also important to recognize that the firewall's configuration, because it is a mechanism for

enforcing policy, imposes its policy on everything behind it. Administrators for firewalls

managing the connectivity for a large number of hosts therefore have a heavy responsibility.

Question:

Why would I want a firewall?

Answer:

The Internet, like any other society, is plagued with the kind of jerks who enjoy

the electronic equivalent of writing on other people's walls with spraypaint, tearing their mailboxes off, or just sitting in the street blowing their car horns. Some people try to get real

work done over the Internet, and others have sensitive or proprietary data they must protect.

Usually, a firewall's purpose is to keep the jerks out of your network while still letting you get

your job done. Many traditional-style corporations and data centers have computing security

policies and practices that must be followed. In a case where a company's policies dictate how

data must be protected, a firewall is very important, since it is the embodiment of the corporate

policy. Frequently, the hardest part of hooking to the Internet, if you're a large company, is not

justifying the expense or effort, but convincing management that it's safe to do so. A firewall

provides not only real security--it often plays an important role as a security blanket for management.

Question:

What can a firewall protect against?

Answer:

Some firewalls permit only email traffic through them, thereby protecting the network against any attacks other than attacks against the email service. Other firewalls provide

less strict protections, and block services that are known to be problems. Generally, firewalls are

configured to protect against unauthenticated interactive logins from the "outside" world. This,

more than anything, helps prevent vandals from logging into machines on your network. More

elaborate firewalls block traffic from the outside to the inside, but permit users on the inside to

communicate freely with the outside. The firewall can protect you against any type of network-

borne attack if you unplug it. Firewalls are also important since they can provide a single choke

point" where security and audit can be imposed. Unlike in a situation where a computer system is

being attacked by someone dialing in with a modem, the firewall can act as an effective`phone

tap" and tracing tool. Firewalls provide an important logging and auditing function; often they

provide summaries to the administrator about what kinds and amount of traffic passed through it,

how many attempts there were to break into it, etc. Because of this, firewall logs are critically

important data. They can be used as evidence in a court of law in most countries. You should

safeguard, analyze and protect your firewall logs accordingly. This is an important point:

providing this ``choke point" can serve the same purpose on your network as a guarded gate can

for your site's physical premises. That means anytime you have a change in ``zones" or levels of

sensitivity, such a checkpoint is appropriate. A company rarely has only an outside gate and no

receptionist or security staff to check badges on the way in. If there are layers of security on your site, it's reasonable to expect layers of security on your network.

Question:

What can't a firewall protect against?

Answer:

Firewalls can't protect against attacks that don't go through the firewall. Many

corporations that connect to the Internet are very concerned about proprietary data leaking out of

the company through that route. Unfortunately for those concerned, a magnetic tape, compact

disc, DVD, or USB flash drives can just as effectively be used to export data. Many

organizations that are terrified (at a management level) of Internet connections have no coherent

policy about how dial-in access via modems should be protected. It's silly to build a six-foot thick steel door when you live in a wooden house, but there are a lot of organizations out there

buying expensive firewalls and neglecting the numerous other back-doors into their network. For

a firewall to work, it must be a part of a consistent overall organizational security architecture.

Firewall policies must be realistic and reflect the level of security in the entire network. For example, a site with top secret or classified data doesn't need a firewall at all: they shouldn't be

hooking up to the Internet in the first place, or the systems with the really secret data should be

isolated from the rest of the corporate network. Lost or stolen PDAs, laptops, cell phones, USB

keys, external hard drives, CDs, DVDs, etc. For protection against this type of data loss, you will

need a good policy, encryption, and some sort of enterprise auditing/enforcement. Places that

really care about Intellectual Property (IP) and data loss prevention use USB firewalling technology on their desktops and systems in public areas. The details are outside the scope of

this FAQ. Badly written, poorly thought out, or non-existent organizational policy. A firewall is

the end extension of an organization's security policy. If that policy is ill-informed, poorly formed, or not formed at all, then the state of the firewall is likely to be similar. Executive buy-in is key to good security practice, as is the complete and unbiased enforcement of your policies.

Firewalls can't protect against political exceptions to the policy, so these must be documented

and kept at a minimum. Another thing a firewall can't really protect you against is traitors or

idiots inside your network. While an industrial spy might export information through your firewall, he's just as likely to export it through a telephone, FAX machine, or Compact Disc. CDs

are a far more likely means for information to leak from your organization than a firewall.

Firewalls also cannot protect you against stupidity. Users who reveal sensitive information over

the telephone are good targets for social engineering; an attacker may be able to break into your

network by completely bypassing your firewall, if he can find a "helpful" employee inside who

can be fooled into giving access to a modem pool or desktop through a "remote support" type portal. Before deciding this isn't a problem in your organization, ask yourself how much trouble

a contractor has getting logged into the network or how much difficulty a user who forgot his

password has getting it reset. If the people on the help desk believe that every call is internal, you have a problem that can't be fixed by tightening controls on the firewalls. Firewalls can't protect

against tunneling over most application protocols to trojaned or poorly written clients. There are

no magic bullets and a firewall is not an excuse to not implement software controls on internal

networks or ignore host security on servers. Tunneling "bad" things over HTTP, SMTP, and

other protocols is quite simple and trivially demonstrated. Security isn't "fire and forget". Lastly, firewalls can't protect against bad things being allowed through them. For instance, many Trojan

Horses use the Internet Relay Chat (IRC) protocol to allow an attacker to control a compromised

internal host from a public IRC server. If you allow any internal system to connect to any external system, then your firewall will provide no protection from this vector of attack.

Question:

How is Network Performance Measured?

Answer:

The performance or "speed" of a computer network is normally measured in units of bits per second (bps). This quantity can represent either an actual data rate or a theoretical

limit to available network bandwidth. The related units of Kbps, Mbps, Gbps represent increasingly larger multiples of bps.

Question:

What Is a Network Name?

Answer:

A network name is a string that computing devices use to identify a specific computer network. Network names are typically different from names of individual computers or

the addresses computers use to identify each other.

Question:

What is an intranet?

Answer:

Intranet is the generic term for a collection of private computer networks within an organization. An intranet uses network technologies as a tool to facilitate communication between people or workgroups to improve the data sharing capability and overall knowledge base of an organization's employees.

Question:

How fast can a normal Null Modem cable transfer files?

Answer:

A null modem cable connects to two standard serial ports for networking two computers together. Null modem cables enable direct data transfer with a minimum of setup required. A null modem cable differs from ordinary serial cables the same way as Ethernet crossover cables differ from ordinary Ethernet cables. Null modem cables reverse the transmit and receive lines on end to enable direct two-way communication. A null modem cable for PCs ordinarily follows the RS-232 standard and uses the same serial ports as RS-232 cables. An RS-232 null modem cable transfers data at the rate of 115 Kbps. The fastest null modem cable, based on RS-422, supports up to 450 Kbps. Today, null modem cables are used primarily by engineers. USB keys, Ethernet crossover cables, and general purpose network routers have effectively made the null modem cable obsolete.

Question:

Who invented the IP - the Internet Protocol?

Answer:

No single person or organization created the modern Internet, including Al Gore, Lyndon Johnson, or any other individual. Instead, multiple people developed the key technologies that later grew to become the Internet: Email - Long before the World Wide Web, email was the dominant communication method on the Internet. Ray Tomlinson developed in 1971 the first email system that worked over the early Internet. Ethernet - The physical communication technology underlying the Internet, Ethernet was created by Robert Metcalfe and

David Boggs in 1973. TCP/IP - In May, 1974, the Institute of Electrical and Electronic Engineers

(IEEE) published a paper titled "A Protocol for Packet Network Interconnection." The paper's authors - Vinton Cerf and Robert Kahn - described a protocol called TCP that incorporated both

connection-oriented and datagram services. This protocol later

Question:

How did the Internet get started?

Answer:

(A) A 1969 U.S. Department of Defense study lead to the deployment of an experimental packet-switched network (the ARPANET) that eventually evolved into the Internet.

The military theorized that a distributed data network would be more fault-tolerant than a telephone network, which could be disabled simply by attacking its central office

Question:

I have two or more computers. How do I connect them to share files and printers?

Answer:

Popular Network Types Ethernet: 10/100Mbps Home PNA 2.0 (Phone Line):

10Mbps Wireless 802.11b: 11 Mbps Wireless HomeRF 2.0: 10Mbps When choosing a network

type (topology) for your home four things should be considered. Cost, Expandability, Location

of Your PC's and speed. This article does not explain every networking type available, but each

of the following network types offer a solution that is suitable for the home, has wide industry

support and offers a good value for the money. Ethernet: The most popular network type for both

home and business is Ethernet. Ethernet is fastest of the network types and can be the least expensive. Ethernet networks are very stable and your network speed will never fluctuate or be

prone to interference like other network types are. Ethernet requires special cables running from

each computer. These cables are plugged into a central 'ethernet hub' or 'switch'. If your computers are far apart, running cable in an existing home may be difficult. Pre-made ethernet

cables come in sizes ranging from 3 feet to 50 feet. Ethernet adapters come in many shapes and

sizes, but PCI Ethernet cards are both the fastest and least expensive. Ethernet can operate at

10Mbps or 100Mbps. When shopping for ethernet equipment, be sure to look for 100Mbps or

10/100Mbps equipment. 10Mbps equipment is older and slower than the 10/100Mbps equipment

and the price difference has become negligible. Note, older 10Mbps equipment will work just

fine on 10/100Mbps networks. If only two computers are being connected, a single Crossover cable can be used instead of the standard cables + switch method.

Question:

How can one disable or enable Simple File Sharing in Windows XP?

Answer:

Simple File Sharing permits controlling both sharing and NTFS permissions at the folder level. Network access is through a guest account. Windows XP Home Edition always

has Simple File Sharing enabled. By default, it is turned off in Windows Professional when it

joins a workgroup. Classic file sharing is used when Windows XP Pro joins a domain. Windows

XP Home cannot join a domain. Simple File Sharing can be disabled in Windows XP Pro when

it is a member of a workgroup. When it is disabled security can be controlled for individual user

accounts. To disable Simple File Sharing, click start, double-click My Computer, Tools, Folder

Options, View tab, at the bottom of the list uncheck Use simple file sharing (Recommended).

Question:

How do I check my network IP address in Windows XP (and in Windows 2000/NT)?

Answer:

Open a DOS windows in Windows XP (Start, All Programs, Accessories,

Command Prompt). Enter at the command prompt enter ipconfig, e.g... C:and

SettingsF. Byard>ipconfig Windows IP Configuration Ethernet adapter Local Area

Connection: Connection-specific DNS Suffix . : IP Address. : 192.168.123.197

**Subnet Mask : 255.255.255.0 Default Gateway : 192.168.123.254 To
obtain the syntax for ipconfig, enter ipconfig /?. C:and SettingsF.**

**Byard>ipconfig /? USAGE: ipconfig [/? | /all | /renew [adapter] | /release [adapter] |
/flushdns |**

**/displaydns | /registerdns | /showclassid adapter | /setclassid adapter [classid]] where
adapter**

**Connection name (wildcard characters * and ? allowed, see examples) Options: /? Display
this**

**help message /all Display full configuration information. /release Release the IP address for
the**

**specified adapter. /renew Renew the IP address for the specified adapter. /flushdns Purges
the**

DNS Resolver cache. /registerdns Refreshes all DHCP leases and re-registers DNS names

**/displaydns Display the contents of the DNS Resolver Cache. /showclassid Displays all the
dhcp**

**class IDs allowed for adapter. /setclassid Modifies the dhcp class id. The default is to
display**

**only the IP address, subnet mask and default gateway for each adapter bound to TCP/IP.
For**

**Release and Renew, if no adapter name is specified, then the IP address leases for all
adapters**

**bound to TCP/IP will be released or renewed. For Setclassid, if no ClassId is specified, then
the**

**ClassId is removed. Examples: > ipconfig ... Show information. > ipconfig /all ... Show
detailed information > ipconfig /renew ... renew all adapters > ipconfig /renew EL* ...
renew any**

**connection that has its name starting with EL > ipconfig /release *Con* ... release all
matching**

connections, eg. "Local Area Connection 1" or "Local Area Connection 2" If your PC is connected to the a device such as broadband router or cable MODEM with a DHCP server, you

are using automatic IP addresses, and you want to test it, config /release and ipconfig /renew to

release and renew the IP address lease. You should see it change, however, the new IP address

may be the same as the old one.

Question:

What is DHCP?

Answer:

DHCP = Dynamic Host Configuration Protocol is an Internet protocol. It resides in a DHCP server and clients that use the server. Simply put, a DHCP server supplies Internet

Protocol (IP) addresses when requested by client computers on a TCP/IP network that have

TCP/IP configured to obtain their IP addresses automatically. A DHCP server is configured to

use a range of IP addresses known as its scope. It automatically and dynamically manages the

allocation of IP addresses within its scope. IP addresses are assigned to clients under a lease arrangement that can be set for to expire after a given time.

Question:

When you have a cable MODEM from an ISP is the DHCP sitting at the ISP site?

Answer:

Yes, one of them is. A DOCSIS (Data Over Cable Service Interface

Specification) cable MODEM also has a DHCP server which supplies a local IP address that is

used to connect to the ISP DHCP server to obtain an IP for the Internet. Is DHCP part of

the Windows 2000 Server operating system? A DHCP server is included as a service with

Windows NT/2000 Server. Windows 2000, 98 SE/Me, etc. Internet Connection Server (ICS) software includes a DHCP server.

Question:

Can a Cat 6 modular plug be plugged into Cat 5 and 5e RJ-45 Jacks?

Answer:

Yes. The physical dimensions of Cat 6 RJ-45 plugs and jacks are identical to Cat

5 and 5e plugs and jacks. They are backward compatible with Cat 3, Cat 5, and Cat 5e

plugs and

jacks.

Question:

What is an Ethernet MAC address?

Answer:

MAC = Media Access Control. Each and every Ethernet device interface to the

network media (e.g., network adapter, port on a hub) has a unique MAC address, which is

"burned" into the hardware when it is manufactured. MAC addresses uniquely identify each node in a network at the Media Access Control layer, the lowest network layer, the one that directly

interfaces with the media, such as the actual wires in a twisted-pair Ethernet. In modern

Ethernets the MAC address consists of six bytes which are usually displayed in hexadecimal;

e.g., 00-0A-CC-32-FO-FD The first three bytes (e.g., 00-0A-CC) are the manufacturer's code

and can be used to identify the manufacturer. The last three are the unique station ID or serial

number for the interface. One can determine the MAC address of an operating Network Interface

Card (NIC or network adapter) in Windows 9X/Me with Start, Run, enter winipcfg, and select

the adapter. In Windows NT, 2000, and XP it can be determined by opening a DOS

Window/Prompt (Start, Programs, Accessories...) and typing: C:ipconfig /all The MAC

address/station ID may be printed on the NIC. Many broadband routers can clone a NIC MAC

address. That is, make the Wide Area Network (WAN) Ethernet interface going to a cable or

DSL MODEM look like a NIC in a PC. This is useful in that many MODEMs marry themselves

to a specific MAC address when they are first installed and it can be rather difficult to get them

to marry themselves to a new MAC address. The WAN port MAC address on some routers can

be manually changed (e.g., the SMC7004ABR).